Kashyap Nukala

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.71
Sri Chaitanya	Class XII (Telangana Board of Intermediate Education)	2024	95.60%
FiitJee World School	Class X (Board of Secondary Education TELANGANA STATE, INDIA)	2022	9.80%

WORK EXPERIENCE

Web Developer -- Rekrite (IIT Hyderabad Startup) | *(website)*

- Built and deployed the official website (duckit.in) for Rekrite, an IIT Hyderabad-incubated startup, using Next.js and hosted it on Vercel, ensuring fast load times and scalable, reliable delivery for end users.
- Collaborated with a 10 member cross functional team to evaluate user needs and refine product experience; recognized by CEO & Founder Dr. Mudrika Khandelwal, HoD of MSME, IIT Hyderabad.

Mini Project: Vacancy Diffusion Simulation -- Dept. of MSME, IIT Hyderabad

- Developed a simulation of vacancy diffusion in a 2D lattice structure under Dr. Anuj Goyal, applying Kinetic Monte Carlo (KMC) methods to model atomic-scale defect migration.

PROJECTS

RAGCore -- Production RAG System | Github

- **Built** a production-grade microservices RAG system using Go/Gin, Python/FastAPI, Qdrant, and PostgreSQL for scalable document ingestion, retrieval, inference, and efficient document management.
- **Evaluated** Dense, Hybrid, and Hybrid+Reranker retrieval pipelines; cross-encoder reranking (ms-marco-MiniLM-L-6-v2) improved retrieval accuracy from **77.8% to 88.9%**.
- Achieved **88.9% answer accuracy** using Llama 3.3 70B (Groq) and optimized chunking through a **15**-configuration grid search, selecting **500**-character chunks with **20%** overlap for improved retrieval quality.
- Profiled the end-to-end RAG pipeline at **706 ms** latency (**12 ms** embedding, **15 ms** retrieval, **115 ms** reranking, **564 ms** generation), identifying inference as the primary bottleneck.

FaceLock - Linux Facial Authentication System | Github

- **Developed** a Linux facial authentication system using **FaceNet, MTCNN,** and **MediaPipe** with blink-based liveness detection, multi-frame face verification, and real time **webcam** based **authentication**.
- **Integrated** the authentication pipeline with **Linux PAM** (pam_exec), enabling **secure biometric login** for KDE Plasma with password fallback while preserving existing Linux authentication workflows.
- **Built** a modular **CLI** supporting user enrollment, authentication, persistent embedding storage, structured logging, performance benchmarking, configuration management, and diagnostic utilities.

GPT-2 Implementation From Scratch | Github

- **Implemented** a decoder-only Transformer (**GPT-2 architecture**) from scratch in PyTorch, building multi-head self-attention, causal masking, residual connections, and layer normalization without high-level Transformer libraries.
- Loaded pretrained **GPT-2 (355M parameter)** OpenAI weights into the custom implementation, writing checkpoint-to-state-dict conversion code and **validating** correctness via coherent text generation before fine-tuning.
- **Fine-tuned** the model for **instruction-following** on an Alpaca-style dataset with response-only loss masking, reducing validation loss from **3.76 to 0.64** over **2** epochs while preserving stable training convergence.
- **Evaluated** generation quality on a held-out **110**-example test set, with qualitative review showing accurate task

completion across factual QA, grammar correction, and figurative-language rewriting.

Physics-Informed Kernel for Ternary Alloy Prediction | Github

- Designed and benchmarked **5 Kernel Ridge Regression kernels** (Linear, RBF, Polynomial, Sign-Flip variants, Exact) for ternary alloy prediction, analytically validating PSD and identifying the invalid Sign-Flip product kernel.
- Demonstrated that the **Polynomial kernel** achieved the best generalization (**RMSE 0.254**) by modeling pairwise alloy interactions, while regional RMSE analysis revealed that the Sign-Flip kernel's strong global RMSE (**0.333**) masked poor performance away from the physical sign-flip boundary.

POSITIONS OF RESPONSIBILITY

Web Domain Head | E-Cell IIT Hyderabad

- Led the web domain team, managing development, deployment, project coordination, and technical strategy for E-Cell's digital platforms, while mentoring members and reviewing code for production deployments.

Web Domain Manager | E-Cell IIT Hyderabad

- Contributed to the development and deployment of **Startup Studio** and **Fetching Fortunes**, building production-grade Next.js applications with Google APIs and managing deployment workflows.

Development Coordinator | Lambda Dev Club, IIT Hyderabad

- Performed QA for the **Mess Management System** serving **6,000+ students** and developed RESTful backend services for the **Key Access Management System** using **Go, PostgreSQL, and Redis**.

SKILLS

Languages & Libraries: Python, Go, JavaScript, SQL, React, Next.js, Tailwind CSS, PyTorch, Transformers, LangChain

Backend & Databases: REST APIs, PostgreSQL, Redis, Qdrant

AI/ML: Retrieval-Augmented Generation (RAG), Neural Networks, Transformers,CNNs

Tools & Platforms: Git, GitHub, Docker, Linux (Ubuntu/Arch), Vercel, Google APIs